1. Linux ⭐ ⭐ ⭐ — Important Interview Questions & Answers

1.What is Linux?
Linux is an open-source operating system widely used for servers, cloud environments, and DevOps tools.
It provides features such as process management, file management, networking, users, permissions, and security.
In DevOps, Linux is commonly used to host applications, configure servers, run automation tools, and troubleshoot production issues.
Simple ga gurthupettuko:
Linux ante **servers and DevOps environments lo commonly use chese operating system**.
Jenkins, Docker, Kubernetes, and many monitoring tools are commonly used in Linux environments.
Eng: Linux is an open-source operating system commonly used for servers, applications, and DevOps environments.

2.What is the Linux File System?
The Linux file system is a hierarchical structure used to organize files and directories.
The root directory / is the starting point of the Linux file system.
All other directories and files are organized under this root directory.
For example, /etc contains configuration files, /var commonly contains logs and application data, and /home contains user home directories.
Simple ga gurthupettuko:
Linux files anni / **root directory nunchi organize avuthayi**.
Different directories have different purposes like configuration, logs, users, and temporary files.
Eng: Linux uses a hierarchical file system starting from the root / directory.

3.What is the difference between a file and a directory?
A file is used to store data, while a directory is used to organize files and other directories.
A file can contain application data, configuration, logs, scripts, or other information.
A directory acts like a folder where files and other directories can be stored.
For example, app.log is a file and /var/log is a directory.
Simple ga gurthupettuko:
File = Data
Directory = Files ni organize cheyyadaniki place
Windows lo folder concept laga Linux lo directory untundi.
Eng: A file stores data, while a directory organizes files and directories.

4.What are Linux users and groups?
A user represents an individual account in Linux.
A group is a collection of users used to manage permissions more easily.
Instead of giving permissions separately to many users, we can assign permissions to a group.
For example, developers can be placed in a developers group and permissions can be given to that group.
Simple ga gurthupettuko:
User = Individual
Group = Users collection
Group use chesthe multiple users permissions easy ga manage cheyyachu.
Eng: Users represent accounts, while groups help manage permissions for multiple users.

5.What are Linux file permissions?
Linux permissions control who can read, write, or execute a file or directory.
The three basic permissions are **read (r), write (w), and execute (x)**.
These permissions help protect files from unauthorized access or modification.
For example, a script needs execute permission if we want to run it directly.
Simple ga gurthupettuko:
r = Read
w = Write
x = Execute
Permissions valla **evaru file ni read/write/execute cheyyalo control cheyyachu**.
Eng: Linux permissions control read, write, and execute access to files and directories.

6.What are Owner, Group, and Others in Linux permissions?
Linux permissions are commonly divided into three categories: **Owner, Group, and Others**.
Owner is the user who owns the file, Group represents users belonging to the file's group, and Others means all remaining users.
Each category can have different read, write, and execute permissions.
For example, the owner may have read/write access while others may have only read access.
Simple ga gurthupettuko:
Owner → File owner
Group → Group members
Others → Remaining users
Eng: Linux permissions can be assigned separately to the owner, group, and others.

7.What is the difference between chmod and chown?
chmod is used to change file or directory permissions.
chown is used to change the owner or group of a file or directory.
So, chmod controls **what users can do**, while chown controls **who owns the file**.
For example, we use chmod to give execute permission and chown to change file ownership.
Simple ga gurthupettuko:
chmod = Permission change
chown = Ownership change
Eng: chmod changes permissions, while chown changes ownership.

8.What is a process in Linux?
A process is a running instance of a program.
When we start an application or command, Linux creates a process to execute it.
Each process normally has a unique Process ID, called a **PID**, which helps us identify and manage it.
For example, when a web server is running, it runs as one or more processes.
Simple ga gurthupettuko:
Program run avuthunte **adi process**.
Process ki **PID** untundi, PID use chesi process ni identify/manage cheyyachu.
Eng: A process is a running instance of a program and has a PID.

9.What is the difference between a process and a service?
A process is a running program, while a service is usually a long-running background application managed by the operating system.
Services are commonly used for applications such as web servers, SSH, databases, and monitoring agents.
A service can create or manage one or more processes.
Simple ga gurthupettuko:
Process = Running program
Service = Background application managed by OS
Example: SSH service background lo run avuthu remote connections accept chestundi.
Eng: A process is a running program, while a service is a managed background application.

10.How do you check running processes in Linux?
We can use commands such as ps and top to check running processes.
ps provides a snapshot of the currently running processes.
top continuously displays processes and their CPU and memory usage, so it is useful for live monitoring.
Simple ga gurthupettuko:
ps = Process snapshot
top = Live process monitoring
High CPU or memory process identify cheyyadaniki top useful.
Eng: I use ps for process information and top for live process monitoring.

11.What is SSH?
SSH stands for **Secure Shell**.
It is a protocol used to securely connect to and manage a remote Linux server over a network.
SSH encrypts the communication between the client and server, making it safer than unsecured remote access methods.
In AWS, SSH is commonly used to connect to Linux EC2 instances.
Simple ga gurthupettuko:
SSH ante **remote Linux server ki secure ga connect avvadaniki use chese protocol**.
AWS lo Linux EC2 ki connect avvadaniki SSH commonly use chestham.
Eng: SSH is a secure protocol used to connect to remote Linux servers.

12.What is the difference between public and private IP?
A public IP can be used to communicate over the public Internet.
A private IP is used for communication within a private network such as a VPC.
Private IPs are commonly used for communication between internal servers, while public IPs are used when Internet connectivity is required.
Simple ga gurthupettuko:
Public IP = Internet communication
Private IP = Internal network communication
AWS lo private EC2 servers generally private IP tho communicate chestayi.
Eng: Public IPs are used for Internet communication, while private IPs are used within private networks.

13.How do you check disk space in Linux?
We can use df to check available and used disk space of file systems.
We can use du to identify how much space specific files or directories are consuming.
This is useful when an application stops because the server disk is full.
Simple ga gurthupettuko:
df = Overall disk space
du = Directory/file usage
Disk full issue vaste first df and du tho usage identify chestham.
Eng: I use df for filesystem usage and du to find which files or directories consume space.

14.How do you check memory usage in Linux?
We can use the free command to check total, used, and available memory.
We can also use top to identify which processes are consuming more memory.
This is useful when an application is slow or the server is running out of memory.
Simple ga gurthupettuko:
free = Memory details
top = Process-wise memory
High memory usage unte **which process consume chestundo** identify chestham.
Eng: I use free and top to check memory usage and identify memory-consuming processes.

15.How do you check whether a Linux service is running?
On systems using systemd, I would use systemctl status to check the service status.
I can also use systemctl to start, stop, restart, or enable services.
This is commonly used while troubleshooting applications and server services.
For example, if a web server is not working, I would first check whether its service is running.
Simple ga gurthupettuko:
systemctl = Service management
Service issue vaste **status → logs → restart** check chestham.
Eng: I use systemctl to check and manage Linux services.